

Ryan Peters

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RESEARCH INTERESTS

Formal methods and verification, with a focus on applying TLA+ and model checking to distributed machine learning training systems. Specification-driven discovery of systems optimizations, distributed systems, and ML systems internals including pipeline and tensor parallelism and sharding protocols.

EDUCATION

Syracuse University, Syracuse, NY
M.S. in Computer Science, expected 2028

Syracuse University, Syracuse, NY
B.S. in Computer Science, [2025]

RESEARCH

M.S. Thesis (in progress): Formal Verification for Distributed ML Training Optimization

Applying TLA+ to specify distributed training protocols, including pipeline parallelism (GPipe, 1F1B, and zero-bubble schedules), ZeRO sharding, tensor parallelism, and ring all-reduce. The central idea is to treat formal specification as a discovery tool that surfaces scheduling slack, redundant synchronization, and refinement opportunities, rather than only verifying correctness after the fact. The methodology centers on four mechanisms: refinement-based design that derives optimized variants from abstract correctness specifications, action decomposition that exposes scheduling freedom, specification weakening that shows certain transformations are safe, and counterexample-guided reduction of over-cautious synchronization. Currently identifying a thesis advisor.

This work builds directly on formal-methods coursework in temporal logic, the mu-calculus, Hennessy-Milner Logic, bisimulation, and process algebra, the logics used to express exactly the safety and liveness properties the specifications target.

TECHNICAL SKILLS

Programming Languages: Python, Racket and Scheme, C and C++, LLVM IR, TypeScript, JavaScript, OCaml (in progress)

Formal Methods and Verification: TLA+, model checking, temporal logic and the mu-calculus, Hennessy-Milner Logic, strong and weak bisimulation, process algebra (CCS and CSP), operational semantics and inference-rule proof systems

Machine Learning and AI: XGBoost and gradient boosting, time-series modeling (lag, rolling, and momentum features, time-series cross-validation), explainable AI (feature-importance analysis), retrieval-augmented generation, multi-agent systems, LLM fine-tuning (LoRA, Unsloth, Ollama), local LLM deployment (llama-cpp-python, quantized GGUF models), Sentence Transformers, spaCy, BLIP, EasyOCR

Data and Systems: pandas, NumPy, ChromaDB, data engineering and multi-frequency time-series alignment, FRED, USDA, and CME market data, Streamlit, Docker, Git, GitHub Actions, Jira, Confluence, Astro, Tailwind CSS, Jupyter, Visual Paradigm, Pixi.js

Software Engineering: object-oriented analysis and design, UML modeling, software requirements specifications, automated validation and QA, team-based development

PROFESSIONAL EXPERIENCE

Support Analyst II, Syracuse University ITS Service Center, Syracuse, NY
[Start year] to Present

- Run front-line and escalated help desk operations supporting students, faculty, and staff across the university.
- Supervise, train, and schedule student technology consultants, and coordinate multi-round hiring cycles.
- Manage Jira ticket queues and maintain support workflows and documentation in Confluence.
- Designed U-Build-It, an appointment-based technology coaching service for students.
- Built an inventory audit workbook and contributed to proactive engagement and service-improvement initiatives.

PROJECTS

Machine Learning and AI

Cross-Market Agricultural Commodity Prediction Engine (Python, XGBoost, Streamlit, llama-cpp-python, pandas)

Lead developer on an explainable-AI dashboard for commodity and livestock markets, built with collaborators for a Syracuse

CCDS project. Implemented a real-time XGBoost forecasting engine with automated feature engineering and time-series cross-validation, a cross-market correlation engine that aligns daily, weekly, and monthly USDA and CME datasets onto a common date index, FRED macro-data integration (corn PPI, crude oil, the Fed funds rate, CPI, and the dollar index), and a privacy-preserving local Llama-3-8B briefing generator. Hardened the system with a 75-check validation suite and an analyst diagnostic notebook.

Multi-Agent Restaurant and Activity Finder (Python, OpenAI, Google Places API, ChromaDB, Streamlit)

Built a chat-first, location-aware recommender on a multi-agent architecture, with an orchestrator coordinating parallel food and activity agents, a scrape-to-vector-store retrieval pipeline with chat memory, and anti-hallucination prompting. Served as integration lead, including secrets management and deployment compatibility.

AI Education Tool (Python, BLIP, EasyOCR)

Contributed memory and performance work to a multimodal tool that converts document pages into audio, including lazy loading of vision and OCR models to reduce startup memory, a redesigned audio interface, step-by-step progress logging, and robust error handling.

Applied LLM Coursework Projects (Python, Streamlit, ChromaDB, Unsloth, Ollama, LoRA)

A series of LLM applications including a RAG pipeline with a vector store, a multi-agent trip planner, a memory-backed chatbot, document and URL summarizers with multi-model switching, and a medical-LLM fine-tuning workflow using Unsloth, Ollama, and LoRA adapters.

Waymo Open Dataset Deep Learning Pipeline (Python, deep learning) (*verify*)

Built a CNN and GRU model for LiDAR-based drive/stop classification for a graduate deep learning course.

Kaggle NVIDIA Nemotron Challenge (Python, QLoRA) (*verify*)

Competed using QLoRA fine-tuning on an A100 GPU.

Systems, Languages, and Web

Scheme-to-LLVM Nanopass Compiler (Racket, LLVM IR)

Designed and built a complete multi-pass compiler that lowers a Scheme subset to LLVM IR and native code, with stages including desugaring, assignment conversion, A-normal form, continuation-passing style, and closure conversion. Also wrote meta-circular and CEK-machine interpreters.

Collaborative Media Review Tool (Vue 3, Pixi.js, Docker)

Built a SyncSketch-style frame-accurate media review and annotation app with WebGL canvas rendering and a containerized setup.

Personal Portfolio Site (Astro, TypeScript, Tailwind CSS, GitHub Actions)

Designed and deployed a portfolio site with interactive elements, including a pipeline-parallelism scheduling visualization, deployed to GitHub Pages through a GitHub Actions pipeline.

SELECTED COURSEWORK

Concurrent Systems and Formal Methods (temporal logic, the mu-calculus, Hennessy-Milner Logic, bisimulation, process algebra), Compiler Construction, Programming Languages, Data Structures and Algorithms, Software Specification and Design, Discrete Mathematics, Natural Language Processing, Building Human-Centered AI Applications, Introduction to Artificial Intelligence.

LEADERSHIP AND SERVICE

- Supervise and train student technology consultants at Syracuse University ITS.
- Designed the U-Build-It student technology coaching service concept.
- Developed an AI workshop proposal for the Blackstone Launchpad. (*verify*)